APM1205 - Linear Models: Summative Assessment 1

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#Introduction and Research Problem

Bike-sharing operators need information about demand to plan bicycle availability throughout the day and across weather conditions. This report analyzes hourly rented-bike counts in Seoul and examines how demand is associated with temperature, humidity, wind speed, and season. Because the dataset is observational, the analysis identifies associations and does not establish causation.

0.1 Research questions and hypotheses

1. Is temperature linearly associated with rented-bike count? The hypotheses are $H_0 : \beta_1 = 0$ and $H_A : \beta_1 \neq 0$.

2. Are temperature, humidity, and wind speed jointly associated with rented-bike demand?
3. Does mean rental demand differ across seasons?
4. Does the relationship between temperature and rental demand differ by season?

1 Part A: Data and Preparation

The analysis uses the Seoul Bike Sharing Demand dataset from the UCI Machine Learning Repository. The response variable is **Rented Bike Count**. Quantitative predictors include temperature, humidity, and wind speed. **Seasons** is categorical. Data types are checked and numeric columns are converted to numeric format. Models use complete observations for their included variables; missing values are not imputed.

The dataset contains **8760 rows** and **14 columns**. There are **0 duplicate rows**.

Table 1: Variable types and missing-value counts

	Variable	Type	Missing
Date	Date	character	0
Rented Bike Count	Rented Bike Count	numeric	0
Hour	Hour	numeric	0
Temperature	Temperature	numeric	0
Humidity	Humidity	numeric	0
Wind speed	Wind speed	numeric	0
Visibility (10m)	Visibility (10m)	numeric	0
Dew point temperature	Dew point temperature	numeric	0
Solar Radiation (MJ/m2)	Solar Radiation (MJ/m2)	numeric	0
Rainfall(mm)	Rainfall(mm)	numeric	0
Snowfall (cm)	Snowfall (cm)	numeric	0
Seasons	Seasons	factor	0
Holiday	Holiday	factor	0
Functioning Day	Functioning Day	factor	0

2 Part B: Exploratory Data Analysis

Table 2: Descriptive statistics for quantitative variables

	Variable	N	Mean	SD	Min	Median	Max
Rented Bike Count	Rented Bike Count	8760	704.60	645.00	0.0	504.50	3556.00
Hour	Hour	8760	11.50	6.92	0.0	11.50	23.00
Temperature	Temperature	8760	12.88	11.94	-17.8	13.70	39.40
Humidity	Humidity	8760	58.23	20.36	0.0	57.00	98.00
Wind speed	Wind speed	8760	1.72	1.04	0.0	1.50	7.40
Visibility (10m)	Visibility (10m)	8760	1436.83	608.30	27.0	1698.00	2000.00

	Variable	N	Mean	SD	Min	Median	Max
Dew point temperature	Dew point temperature	8760	4.07	13.06	-30.6	5.10	27.20
Solar Radiation (MJ/m2)	Solar Radiation (MJ/m2)	8760	0.57	0.87	0.0	0.01	3.52
Rainfall(mm)	Rainfall(mm)	8760	0.15	1.13	0.0	0.00	35.00
Snowfall (cm)	Snowfall (cm)	8760	0.08	0.44	0.0	0.00	8.80

Table 3: Bike rentals by season

Seasons	N	Mean	SD
Autumn	2184	819.60	651.09
Spring	2208	730.03	621.51
Summer	2208	1034.07	690.24
Winter	2160	225.54	150.37

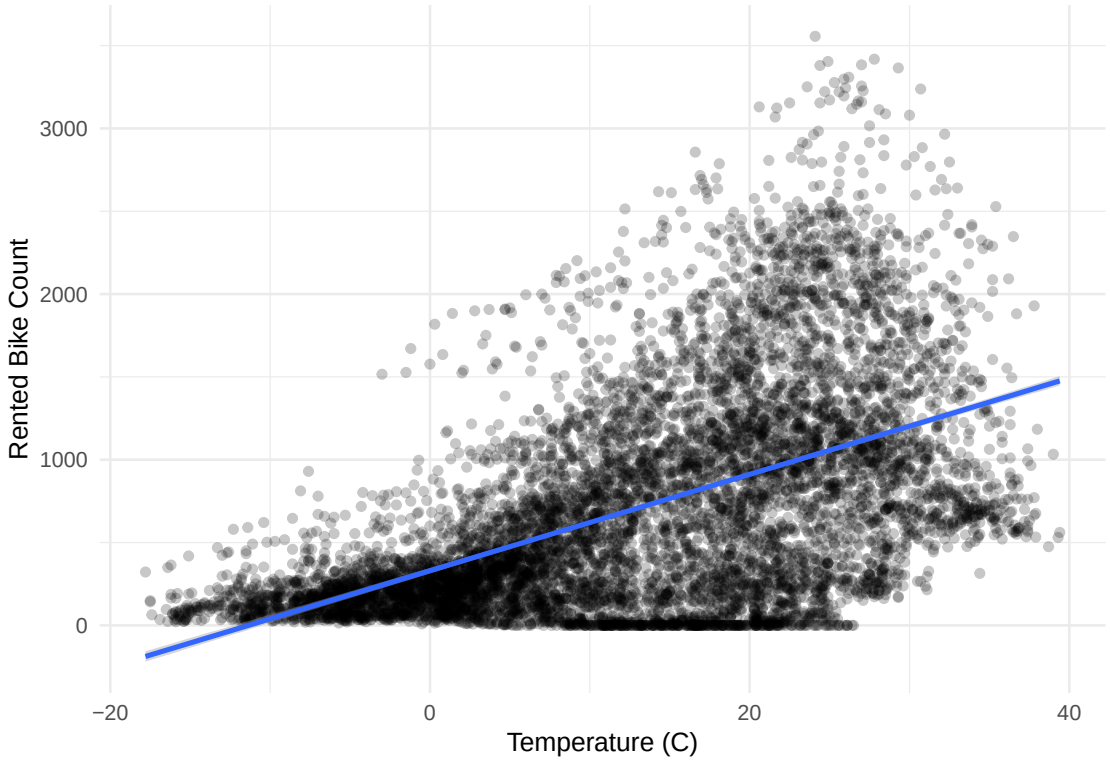


Figure 1: Rented-bike count versus temperature with a fitted linear trend.

The plots help identify trends, differences between seasons, and possible unusual observations. Visual patterns should be interpreted together with the numerical summaries rather than on their own.

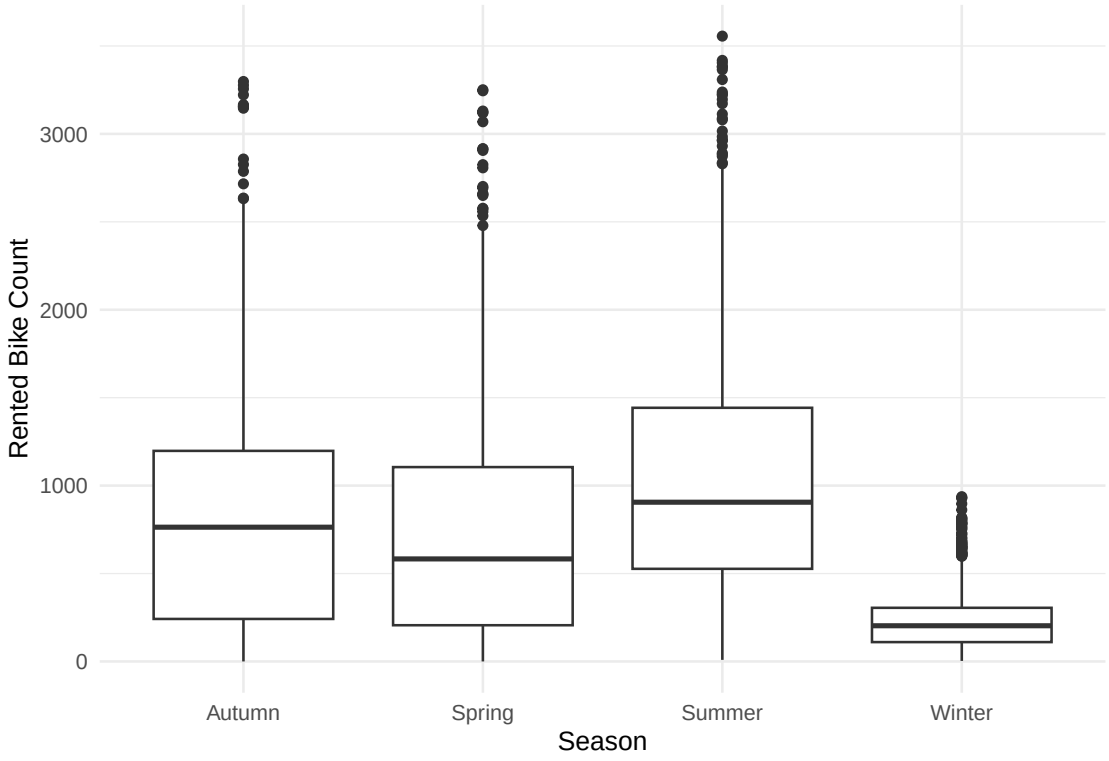


Figure 2: Distribution of rented-bike counts by season.

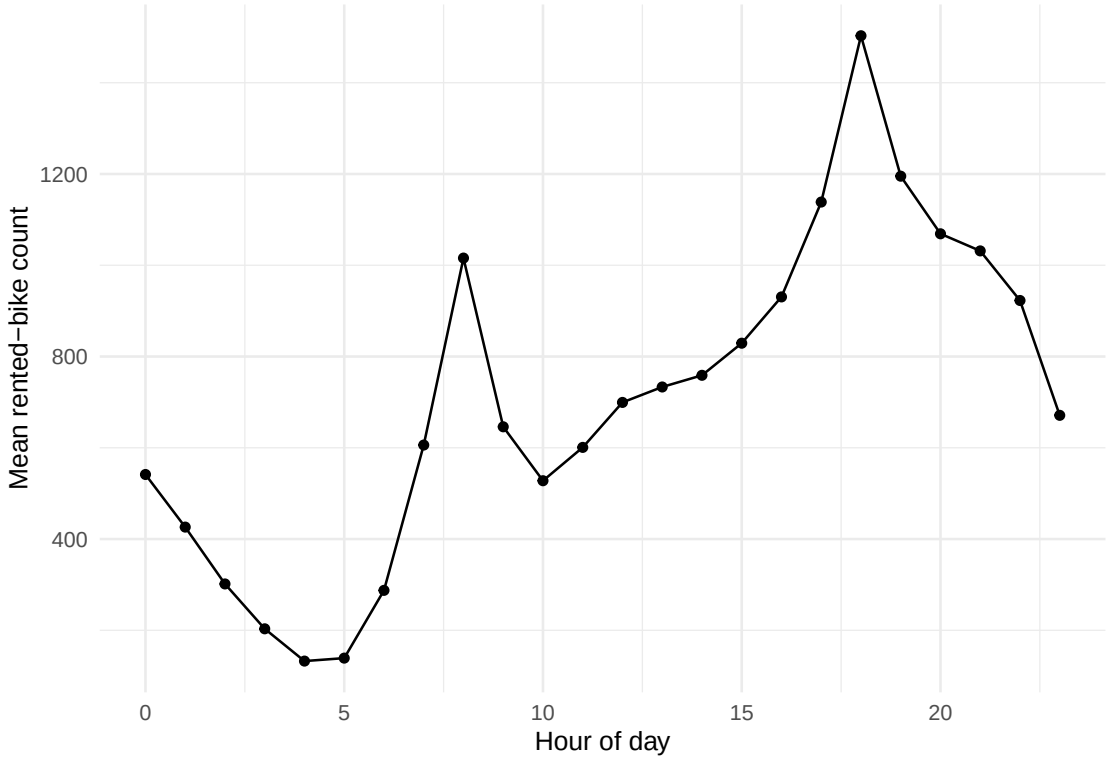


Figure 3: Average bike rentals by hour of day.

3 Part C: Simple Linear Regression

The simple regression model is $Y = \beta_0 + \beta_1 X + \varepsilon$, where Y is rented-bike count and X is temperature.

Table 4: Simple regression coefficients and 95% confidence intervals

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	329.9525	8.5411	38.6313	0	313.2100	346.6950
Temperature	29.0811	0.4862	59.8163	0	28.1281	30.0341

Table 5: Simple regression model fit

r.squared	adj.r.squared	sigma	statistic	p.value
0.29	0.29	543.4984	3577.991	0

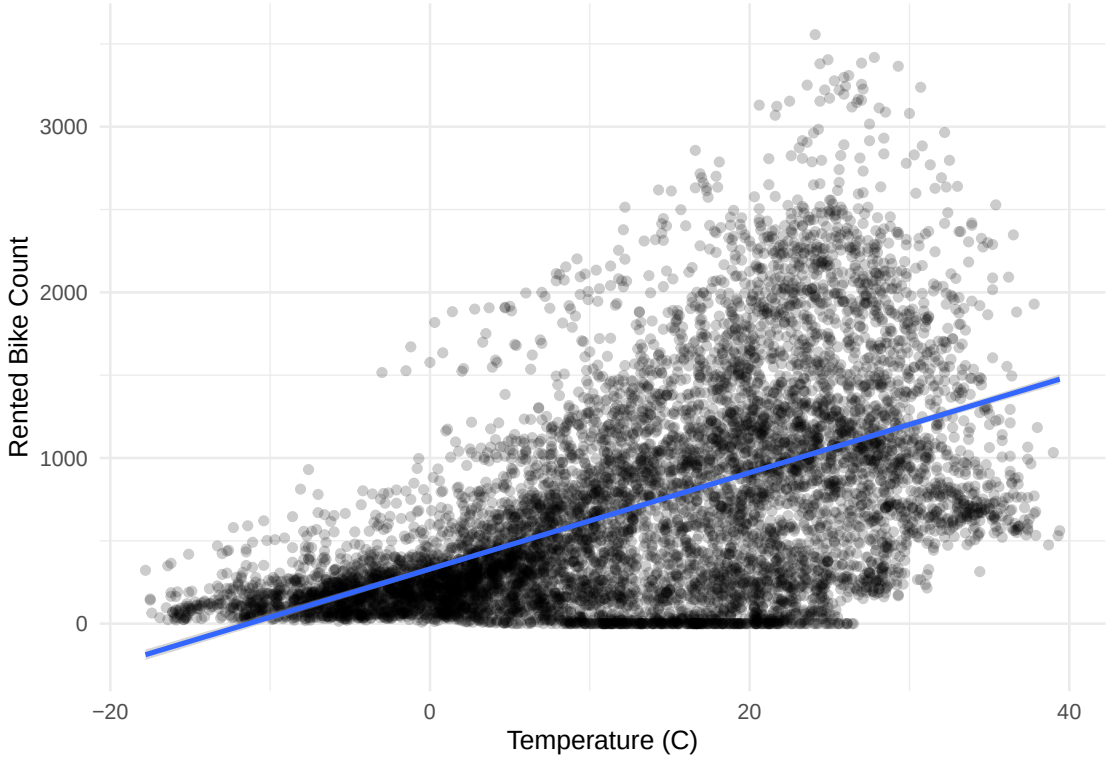


Figure 4: Simple linear regression of rented-bike count on temperature.

The estimated temperature slope is **29.081**. This represents the estimated average change in rented-bike count associated with a one-unit increase in temperature in the unadjusted model. The model R^2 is **0.29**, the proportion of sample variation in rental counts explained by this simple linear model.

4 Part D: Multiple Linear Regression

The multiple regression model includes temperature, humidity, and wind speed. Each coefficient is interpreted while holding the other included predictors constant.

Table 6: Multiple regression coefficients and 95% confidence intervals

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	754.8449	22.6470	33.3309	0	710.4514	799.2384
Temperature	31.5555	0.4619	68.3217	0	30.6502	32.4609
Humidity	-8.7530	0.2875	-30.4403	0	-9.3166	-8.1893
`Wind speed`	30.6583	5.5815	5.4929	0	19.7173	41.5993

Table 7: Multiple regression model fit

r.squared	adj.r.squared	statistic	p.value	df	df.residual
0.3759	0.3757	1757.895	0	3	8756

The model has (R^2=)0.3759 and adjusted (R^2=)0.3757. The overall F-test p-value is **0**. A small p-value provides evidence that at least one predictor has a nonzero slope in this model.

5 Part E: Statistical Inference

The following table gives coefficient estimates, standard errors, p-values, and 95% confidence intervals for the multiple regression model.

Table 8: Coefficient inference for the multiple regression model

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	754.8449	22.6470	33.3309	0	710.4514	799.2384
Temperature	31.5555	0.4619	68.3217	0	30.6502	32.4609
Humidity	-8.7530	0.2875	-30.4403	0	-9.3166	-8.1893
`Wind speed`	30.6583	5.5815	5.4929	0	19.7173	41.5993

For temperature, the estimated coefficient is **31.556**, with a 95% confidence interval from **30.65** to **32.461**. Its p-value is **0**. Statistical significance does not automatically imply that an effect is practically important; the magnitude and context of the estimate should also be considered.

6 Part F: Matrix Form and Least Squares

The linear model can be written as $\mathbf{y} = \mathbf{X}\boldsymbol{\beta} + \boldsymbol{\varepsilon}$. Here, $\mathbf{y}$ is the response vector, $\mathbf{X}$ is the design matrix containing an intercept and predictor columns, $\boldsymbol{\beta}$ is the coefficient vector, and $\boldsymbol{\varepsilon}$ is the error vector. When $\mathbf{X}^T\mathbf{X}$ is invertible, the least-squares estimator is

$$\hat{\boldsymbol{\beta}} = (\mathbf{X}^T\mathbf{X})^{-1}\mathbf{X}^T\mathbf{y}.$$

Table 9: Matrix least-squares estimates compared with lm() on the same subset

Term	Matrix_Estimate	lm_Estimate	Difference
(Intercept)	575.190585	575.190585	0
Temperature	35.651613	35.651613	0
Humidity	-9.290525	-9.290525	0
`Wind speed`	133.420831	133.420831	0

The estimates should agree up to numerical rounding. This connects the matrix formula to the regression procedure used by R's `lm()` function.

7 Part G: Dummy Variables and ANOVA

Season is categorical, so R represents its levels using indicator (dummy) variables. The first factor level is the reference category under the default treatment coding. The intercept is the estimated mean for the reference season, and each other season coefficient is the estimated difference from that reference in the season-only model.

Table 10: Season dummy-variable regression

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	819.5980	12.2683	66.8063	0	795.5493	843.6467
SeasonsSpring	-89.5667	17.3027	-5.1764	0	-123.4842	-55.6493
SeasonsSummer	214.4754	17.3027	12.3955	0	180.5580	248.3928
SeasonsWinter	-594.0568	17.3981	-34.1450	0	-628.1611	-559.9525

Table 11: ANOVA testing mean differences among seasons

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Seasons	3	765709006	255236335.5	776.4678	0
Residuals	8756	2878225356	328714.6	NA	NA

The ANOVA null hypothesis is that the mean rented-bike count is equal across all seasons. The ANOVA p-value is **0**. If this value is below the chosen significance level (commonly 0.05), the data

provide evidence that at least one seasonal mean differs. This overall test does not identify every pair that differs.

8 Part H: Interaction Analysis

The interaction model allows the temperature slope to vary by season: $Y = \beta_0 + \beta_1 T +$ Season effects $+ T \times$ Season $+ \varepsilon$.

Table 12: Temperature-by-season interaction model coefficients

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	405.9878	25.4891	15.9279	0.0000	356.0231	455.9525
Temperature	29.2908	1.6145	18.1424	0.0000	26.1260	32.4556
SeasonsSpring	-360.0734	35.7508	-10.0718	0.0000	-430.1534	-289.9934
SeasonsSummer	-15.2369	70.4175	-0.2164	0.8287	-153.2717	122.7979
SeasonsWinter	-153.7078	28.4507	-5.4026	0.0000	-209.4779	-97.9378
Temperature:SeasonsSpring	23.1453	2.3544	9.8305	0.0000	18.5300	27.7605
Temperature:SeasonsSummer	-5.0901	2.9193	-1.7436	0.0813	-10.8126	0.6325
Temperature:SeasonsWinter	-18.7656	2.6458	-7.0926	0.0000	-23.9520	-13.5792

Table 13: ANOVA table for the interaction model

	Df	Sum Sq	Mean Sq	F value	Pr(>F)
Temperature	1	1056904520	1056904520.2	3724.0660	0
Seasons	3	29334094	9778031.5	34.4535	0
Temperature:Seasons	3	73843862	24614620.6	86.7311	0
Residuals	8752	2483851886	283803.9	NA	NA

The interaction terms test whether the temperature-demand relationship differs by season. Use the coefficient table and interaction plot to describe the direction and size of differences. The overall model ANOVA compares sequential terms; a joint test of all interaction coefficients can be obtained by comparing models with and without the interaction using `anova()`.

Table 14: Joint test of whether adding the temperature-by-season interaction improves fit

Res.Df	RSS	Df	Sum of Sq	F	Pr(>F)
8755	2557695748	NA	NA	NA	NA
8752	2483851886	3	73843862	86.7311	0

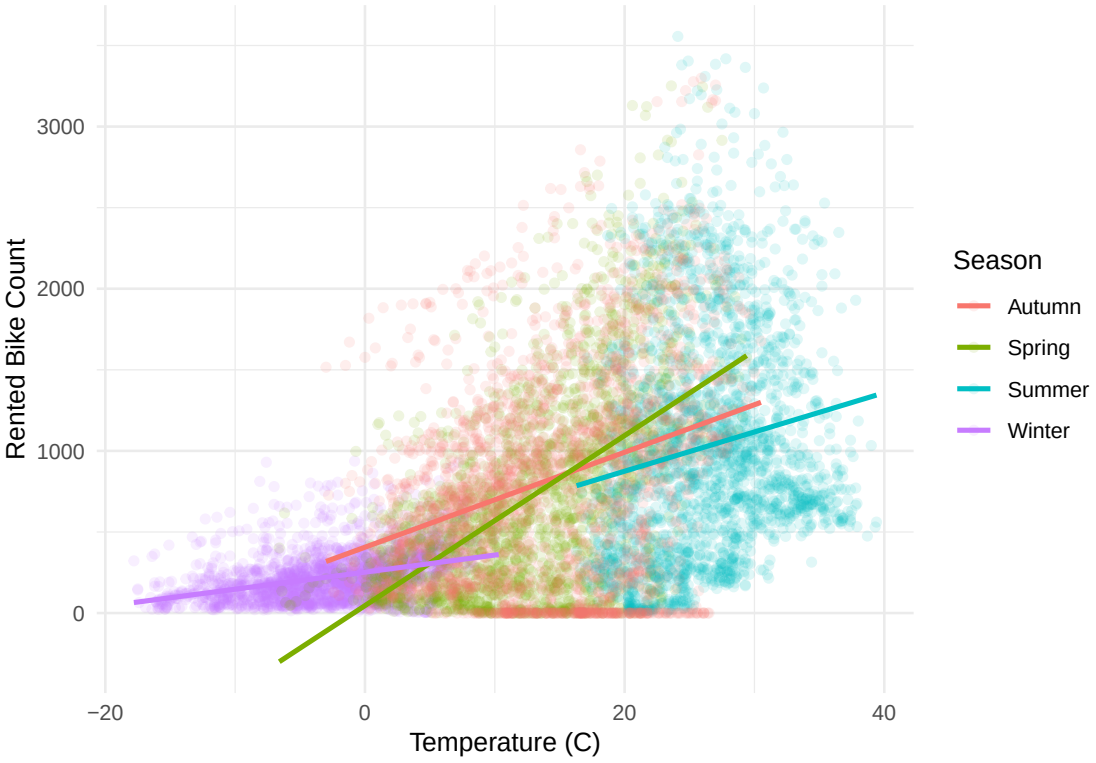


Figure 5: Estimated temperature-demand trends across seasons.

9 Part I: Model Diagnostics

The final model includes temperature, humidity, wind speed, season, and the temperature-by-season interaction.

Table 15: Regression diagnostic tests

Test	Statistic	P_value
Breusch-Pagan test (non-constant variance)	1050.7829	0
Durbin-Watson test (serial correlation)	0.3339	0
RESET test (functional form)	240.7590	0

Table 16: Ten observations with the largest Cook's distances

Row	Cooks_Distance
6859	0.00326
6811	0.00319
6979	0.00305
6667	0.00299
6883	0.00298
6835	0.00289

Row	Cooks_Distance
6691	0.00280
4291	0.00268
4363	0.00245
4339	0.00235

Residual-versus-fitted plots are used to assess linearity and constant variance; the Q-Q plot assesses whether residuals are approximately normal; Cook's distance helps identify potentially influential observations. For the Breusch-Pagan and RESET tests, small p-values suggest possible non-constant variance or functional-form problems. The Durbin-Watson test is relevant to possible serial correlation, especially because these are hourly observations. These diagnostics are indicators, not definitive proof that assumptions hold or fail.

10 Part J: Final Model and Conclusion

The final model's R^2 is **0.4049** and adjusted R^2 is **0.4043**.

Table 17: Final model coefficient estimates and 95% confidence intervals

term	estimate	std.error	statistic	p.value	conf.low	conf.high
(Intercept)	950.8809	31.9411	29.7698	0.00	888.2688	1013.4930
Temperature	25.4820	1.5167	16.8015	0.00	22.5090	28.4550
Humidity	-9.3181	0.2964	-31.4412	0.00	-9.8991	-8.7372
`Wind speed`	40.6682	5.5518	7.3252	0.00	29.7853	51.5511
SeasonsSpring	-331.5809	33.5865	-9.8724	0.00	-397.4184	-265.7433
SeasonsSummer	563.9930	67.7826	8.3206	0.00	431.1233	696.8628
SeasonsWinter	-291.6238	26.9824	-10.8079	0.00	-344.5156	-238.7319
Temperature:SeasonsSpring	19.1411	2.2050	8.6806	0.00	14.8187	23.4635
Temperature:SeasonsSummer	-23.2531	2.7753	-8.3785	0.00	-28.6933	-17.8128
Temperature:SeasonsWinter	-6.4389	2.5006	-2.5749	0.01	-11.3406	-1.5371

Table 18: Final model fit statistics

r.squared	adj.r.squared	statistic	p.value	df	df.residual
0.4049	0.4043	661.5942	0	9	8750

In conclusion, the analysis evaluates how temperature, humidity, wind speed, and season are associated with hourly rented-bike demand in Seoul. The coefficient estimates and confidence intervals describe the estimated relationships, while the model fit statistics summarize how much variation is explained. The interaction analysis assesses whether the temperature relationship differs by season, and the diagnostic plots and tests indicate whether the model's assumptions appear reasonable. The final interpretation should identify the predictors with evidence of association using their p-values

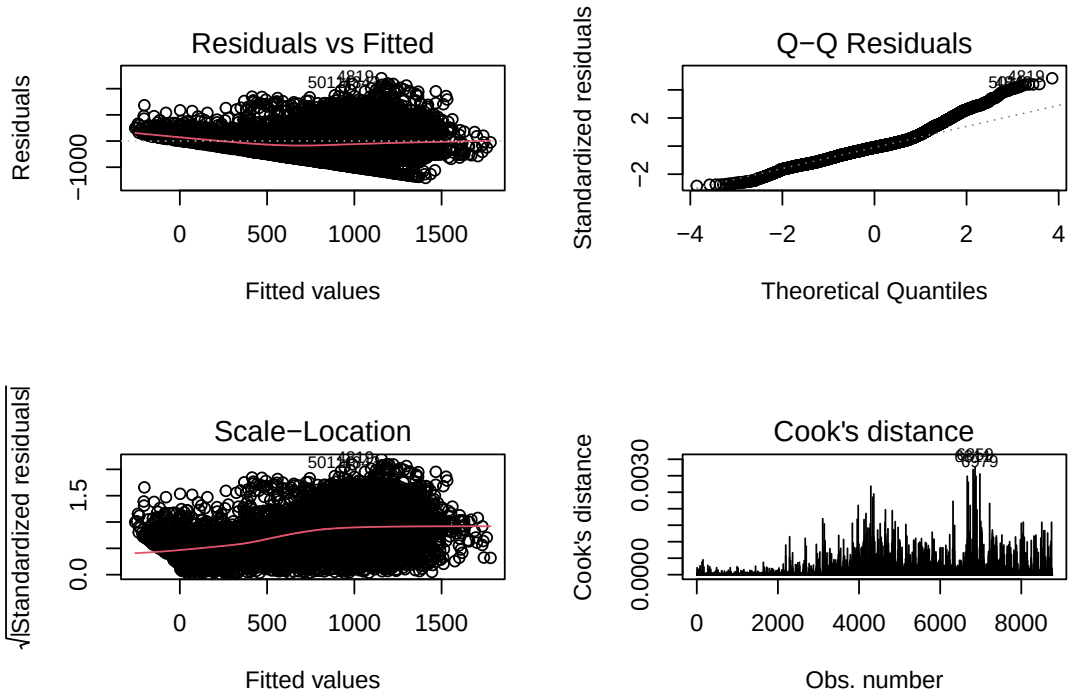


Figure 6: Standard diagnostic plots for the final model.

and confidence intervals, discuss whether the estimated effects are meaningful in the bike-sharing context, and acknowledge any diagnostic concerns. These are observational associations and should not be interpreted as proof of causation.

11 References

UCI Machine Learning Repository. (2020). *Seoul Bike Sharing Demand* [Dataset]. https://doi.org/10.24432/C5F62R. Licensed under CC BY 4.0.

R Core Team. *R: A language and environment for statistical computing*. https://www.r-project.org/